Individual Report

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Supervised by: Prof. / Dr. Alaa Hamdy

Team ID: 2022/05885

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GitHub: https://github.com/Rowlkh

1. Tasks

Machine Learning Development:

- Pre-processing for LSTM, GRU, and RNN models
- Pre-processing for HMM
- Modeling for LSTM, GRU, RNN, and HMM
- Fine-tuning for all four models (LSTM, GRU, RNN, HMM)

System Design:

- System overview diagram
- Complete system overview documentation
- ESP32diagram

PowerPoint Presentation:

- Requirements section
- Functional Requirements
- Use Case Diagram
- Non-Functional Requirements

SRS Documentation:

- Complete Chapter 2
- Chapter 1: Project management and deliverables section (including project time plan, and Gantt chart)
- Chapter 3: Introduction
- Chapter 4: Data design, testing plan, requirement matrix, system deployment, and chapter summary

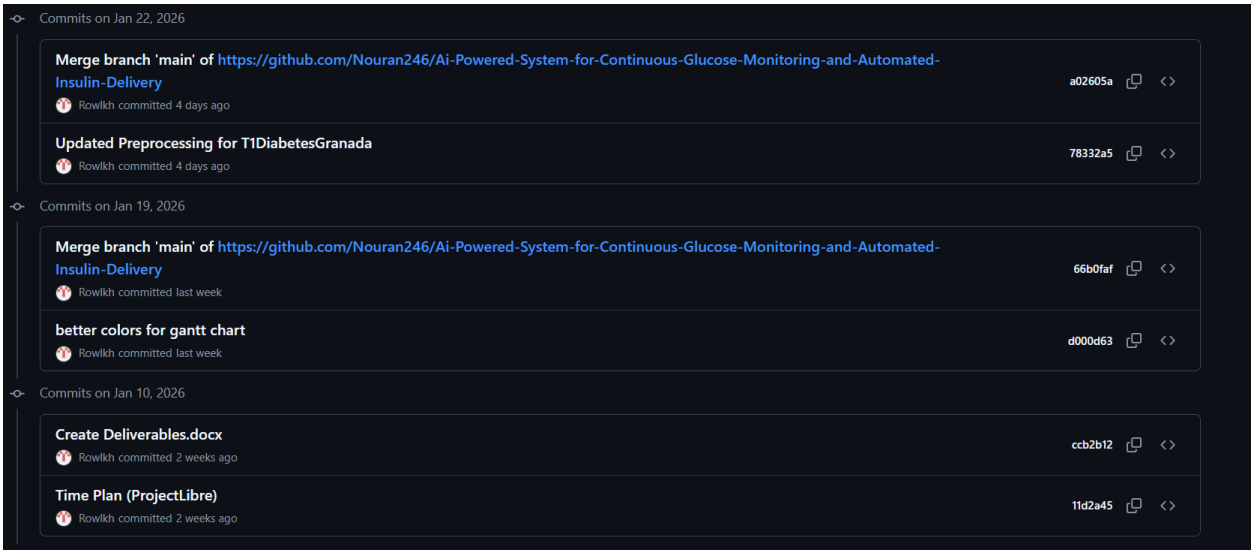
Proposal Document:

- Expected contributions section
- Pre-processing on preliminary dataset
- Literature review - Question 5

Table 1: Plan

Req Code	Description	Start Date	End Date
1	Proposal – Literature Review (Q5)	25/09/2025	09/10/2025
2	Proposal – Expected Contributions	25/09/2025	18/11/2025
3	Proposal – Pre-processing (Dataset)	25/09/2025	18/11/2025
4	Pre-processing for LSTM, GRU, RNN	23/12/2025	31/12/2025
5	Pre-processing for HMM	23/12/2025	31/12/2025
6	Modeling LSTM, GRU, RNN, HMM	01/01/2026	06/01/2026
7	Fine-tuning all models	01/01/2026	06/01/2026
8	System Overview Diagram	08/01/2026	29/01/2026
9	ESP32 Overview Diagram	08/01/2026	29/01/2026
10	SRS – Chapter 2 (Complete)	08/01/2026	29/01/2026
11	SRS – Chapter 1 (Project Management)	08/01/2026	29/01/2026
12	SRS – Chapter 3 (Introduction)	08/01/2026	29/01/2026
13	SRS – Chapter 4 (Data Design, Testing)	08/01/2026	29/01/2026
14	PowerPoint – Requirements & Use Cases	08/01/2026	29/01/2026

2. GitHub



better colors for gantt chart
Rowlkh committed last week — d000d63

Commits on Jan 10, 2026

Create Deliverables.docx
Rowlkh committed 2 weeks ago — ccb2b12

Time Plan (ProjectLibre)
Rowlkh committed 2 weeks ago — 11d2a45

Commits on Jan 7, 2026

Add files via upload
Rowlkh authored 3 weeks ago — Verified — c050e4a

Commits on Jan 6, 2026

GRU using multipatient
Rowlkh committed 3 weeks ago — d183de4

Preprocessing (single/multi patient) + GRU/LSTM/RNN Modeling
Rowlkh committed 3 weeks ago — 45b24a2

Commits on Nov 8, 2025

few updates
Rowlkh committed on Nov 8, 2025 — 9e69ac3

Ai-Powered-System-for-Continuous-Glucose-Monitoring-and-Automated-Insulin-Delivery / Diagrams / **Timeplan & Delieverables** /

Add file

Rowlkh better colors for gantt chart — d000d63 · last week — History

Name	Last commit message	Last commit date
..		
Deliverables.docx	Create Deliverables.docx	2 weeks ago
GP.pod	Time Plan (ProjectLibre)	2 weeks ago
GanttChart.pdf	better colors for gantt chart	last week
Timeline.png	better colors for gantt chart	last week

Ai-Powered-System-for-Continuous-Glucose-Monitoring-and-Automated-Insulin-Delivery / Preprocessing / **diabetes dataset (Time-Series) - Roaa** /

Add file

Rowlkh few updates — 9e69ac3 · 2 months ago — History

Name	Last commit message	Last commit date
..		
cleaned_datasets	few updates	2 months ago
merged_across_patients	Preprocessing	3 months ago
scripts	few updates	2 months ago

Ai-Powered-System-for-Continuous-Glucose-Monitoring-and-Automated-Insulin-Delivery / Preprocessing / T1DiabetesGranada /

Add file

Rowlkh Updated Preprocessing for T1DiabetesGranada 78332a5 · 4 days ago History

Name	Last commit message	Last commit date
..		
Modeling	Add files via upload	3 weeks ago
Preprocessing	Updated Preprocessing for T1DiabetesGranada	4 days ago